

Module 9: Risk in E Commerce

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Risk of E–Commerce

E–Commerce involves buying and selling goods and services through electronic networks such as the internet. Since business transactions are performed online, e–commerce systems face several security and operational risks. These risks may affect customers, businesses, payment systems, and confidential information.

E–commerce security focuses on protecting online transactions, customer data, communication networks, and business systems from unauthorized access, fraud, and cyber-attacks.

Types of Risks in E–Commerce

1) Security Risks: Unauthorized users may access confidential business or customer information. **Examples**

- Hacking
- Data theft
- Malware attacks
- Identity theft

2) Financial Risks: Financial loss may occur because of fraudulent transactions or payment failures. **Examples**

- Credit card fraud
- Fake transactions
- Online payment scams

3) Operational Risks: Technical failures can interrupt online business operations.

Examples

- Server failure
- Software errors
- Network breakdown

4) Legal Risks: Businesses may violate laws related to privacy, taxation, or intellectual property. **Examples**

- Copyright violation
- Data privacy issues
- Non-compliance with regulations

5) Reputation Risks: Security breaches may reduce customer trust and damage company reputation. **Examples**

- Leakage of customer information
- Poor service quality
- Website downtime

6) Privacy Risks: Personal and financial information of customers may be misused.

Examples

- Unauthorized data sharing
- Tracking customer activities without permission

Security for E–Commerce

E–commerce security refers to the methods and technologies used to protect online transactions, customer information, communication systems, and electronic payments. **Objectives**

1. **Confidentiality** – Protects sensitive information from unauthorized access.
2. **Integrity** – Ensures data is not altered during transmission.
3. **Authentication** – Verifies identity of users and systems.
4. **Non-Repudiation** – Prevents denial of completed transactions.
5. **Availability** – Ensures systems and services remain accessible.

Security Standards in E–Commerce

Security standards are rules and protocols designed to protect electronic transactions and communication.

1) SSL (Secure Socket Layer)

- Encrypts data transmitted between browser and server.
- Protects passwords, payment details, and customer information.
- Indicated by HTTPS in websites.

2) TLS (Transport Layer Security)

- Advanced version of SSL.
- Provides secure communication over networks.

3) SET (Secure Electronic Transaction)

- Security standard developed for safe credit card transactions.
- Uses encryption and digital certificates.

4) PCI DSS (Payment Card Industry Data Security Standard)

- Security standard for organizations handling card payments.
- Protects cardholder information from fraud.

Firewall

A firewall is a network security system that monitors and controls incoming and outgoing network traffic based on security rules. It acts as a protective barrier between a trusted internal network and untrusted external networks such as the internet. **Functions of Firewall**

1. Blocks unauthorized access.
2. Monitors network traffic.
3. Prevents hacking attempts.
4. Protects servers and databases.
5. Filters malicious data packets.

Types of Firewalls

1. **Packet Filtering Firewall:** Checks data packets and allows or blocks them.
2. **Proxy Firewall:** Acts as an intermediary between users and the internet.
3. **Stateful Inspection Firewall:** Monitors active connections and network states.
4. **Application Firewall:** Protects specific applications such as web servers.

Advantages of Firewall

1. Improves network security.
2. Prevents unauthorized access.
3. Reduces cyber-attacks.
4. Monitors internet activities.
5. Protects confidential business data.

Cryptography

Cryptography is the process of converting readable information into coded form to protect data from unauthorized access. The original message is called **plaintext**, and the coded message is called **ciphertext**. **Objectives of Cryptography**

1. Data confidentiality
2. Data integrity
3. Authentication
4. Non-repudiation

Types of Cryptography

1) Symmetric Key Cryptography:

Same key is used for encryption and decryption. Faster and simpler method.

- **Examples** DES, AES
- **Advantages** High speed, Suitable for large data transfer
- **Disadvantages:** Key distribution problem

2) Asymmetric Key Cryptography: Uses two keys: public key and private key. Public key encrypts data, private key decrypts data.

Examples: RSA, ECC

Advantages: More secure, Supports digital signatures

Disadvantages: Slower than symmetric encryption

Key Management

Key management refers to the generation, storage, distribution, updating, and protection of cryptographic keys. Proper key management is essential for maintaining security in e-commerce systems.

Functions of Key Management

1. Key generation
2. Key distribution
3. Key storage
4. Key backup and recovery
5. Key replacement and destruction

Characteristics of Strong Passwords

1. Combination of letters, numbers, and symbols.
2. Difficult to guess.
3. Regularly updated.
4. Unique for each account.

Importance of Key Management

1. Protects encrypted information.
2. Prevents unauthorized access.
3. Maintains secure communication.
4. Reduces risk of data leakage.

Types of Password Systems

- 1) **Static Password:** Fixed password chosen by the user.
- 2) **One-Time Password (OTP):** Temporary password valid for a single session.
- 3) **Biometric Authentication:** Uses fingerprint, face recognition, or iris scan.
- 4) **Multi-Factor Authentication (MFA):** Uses more than one verification method.

Password Systems

Password systems are authentication mechanisms used to verify the identity of users before granting access to systems or accounts.

Advantages of Password Systems

1. Prevents unauthorized access.
2. Protects user accounts and transactions.
3. Enhances online security.

Digital Certificates

A digital certificate is an electronic document used to verify the identity of a person, website, or organization in online communication. It is issued by a trusted authority called a **Certificate Authority (CA)**.

Contents of Digital Certificate

1. Owner's name
2. Public key
3. Certificate serial number
4. Expiry date
5. Digital signature of CA

2. Secure data communication.
3. Prevention of fake websites.
4. Support for encrypted transactions.

Functions of Digital Certificates

1. Authentication of websites and users.

Advantages of Digital Certificates

1. Improves trust in e-commerce websites.
2. Protects customer information.
3. Ensures secure online transactions.

Digital Signatures

A digital signature is an electronic method used to verify authenticity and integrity of electronic documents or messages. It is created using cryptographic techniques and private keys.

Working of Digital Signature

1. Sender creates message.
2. Message is converted into hash value.
3. Hash value is encrypted using sender's private key.
4. Receiver decrypts using sender's public key.
5. If values match, authenticity is verified.

Advantages of Digital Signatures

1. Reduces document forgery.
2. Ensures secure communication.
3. Increases trust in online transactions.
4. Supports paperless business operations.
5. Provides legal validity to electronic documents.

Functions of Digital Signature

1. Authentication of sender.
2. Ensures data integrity.
3. Provides non-repudiation.
4. Secures electronic transactions.

Importance of Security in E-Commerce

1. Protects customer and business information.
2. Ensures safe online transactions.
3. Prevents fraud and cyber-attacks.
4. Builds customer trust and confidence.
5. Supports growth of digital business and online banking.